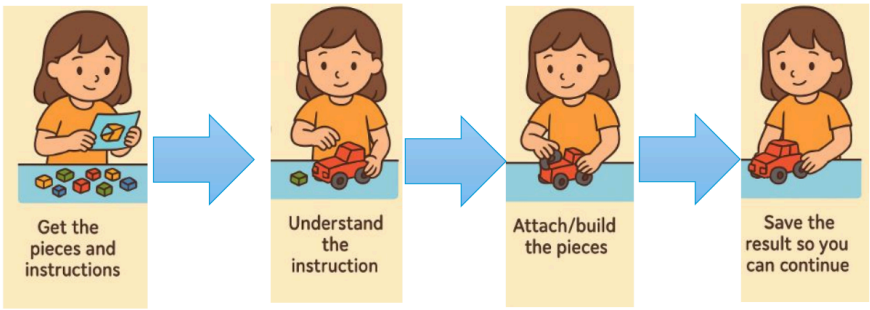
CPU (Central Processing Unit)

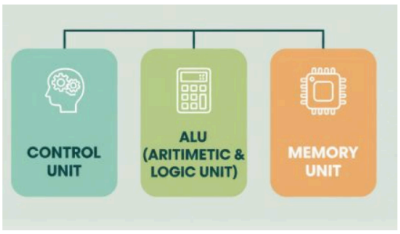
✔ **How it works?**



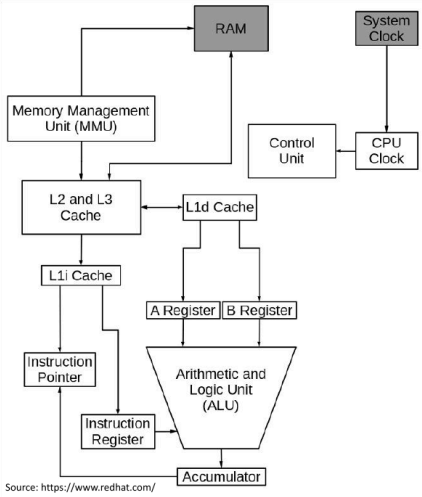
FETCH	DECODE	EXECUTE	STORE
The CPU collects instructions from the system's memory.	It changes those instructions into easy-to-follow steps.	The CPU performs the action, such as solving math or moving information.	The output is kept in memory so it can be used later when required.

CPU (Central Processing Unit)

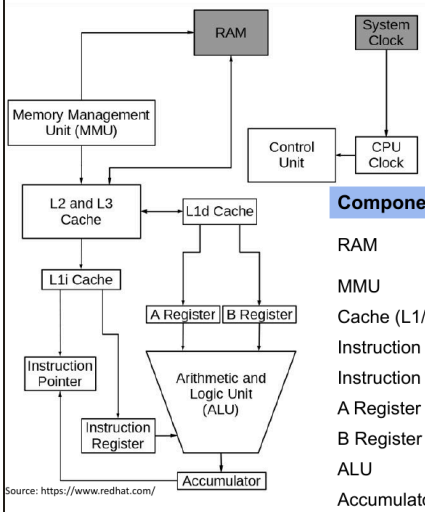
✔ **Parts of CPU**



Control Unit (CU): Directs the flow of instructions and data within the computer.
Arithmetic Logic Unit (ALU): Performs calculations and logical operations.
Registers: Small, high-speed storage locations for temporarily holding data.
Cache Memory: Extremely fast memory that provides rapid access to frequently used data and instructions.
Buses: Pathways that carry data and signals between various parts of the computer.



Working Procedure

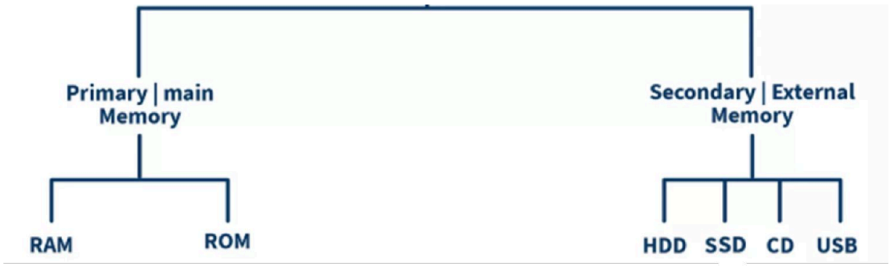


Let's add **two numbers** , **example** : 5 + 3.

Component	Action	Value / Example
RAM	Stores instructions and data	`LOAD A,5`, `LOAD B,3`, `ADD A,B`
MMU	Maps virtual → physical addresses	`0x01 → 0x101`
Cache (L1/L2/L3)	Holds frequently used instructions/data	`LOAD A,5`
Instruction Pointer	Points to next instruction	`0x01 → 0x02`
Instruction Register	Holds current instruction	`LOAD A,5`
A Register	Holds first operand	5
B Register	Holds second operand	3
ALU	Performs arithmetic/logic	5 + 3 = 8
Accumulator	Stores result	8

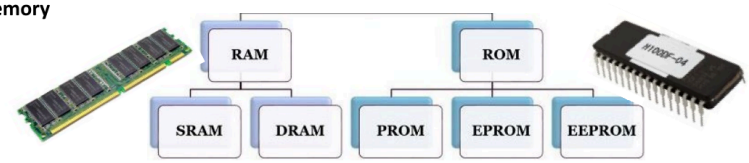
Memory Unit

Types of Computer Memory



Memory Unit

Primary Memory



- ✔ **RAM** stands for Random Access Memory, and it stores data temporarily while the computer is in use.
- ✔ It helps the computer remember what you are doing at the moment.
- ✔ Volatile (loses data when powered off)

How does RAM work?

1. You turn on your computer, and important files load into RAM.
2. When you open an app, RAM stores it for quick access and sends data to the CPU for processing.
3. As you open more apps, RAM divides its space among them.
4. Once you close the app or shut down the computer, everything in RAM is erased.

Memory Unit

Primary Memory (RAM)

Static RAM (SRAM)

- ✔ SRAM is a high-speed type of memory that doesn't need constant refreshing.
  - Stores data as long as power is supplied
  - Operates very fast
  - More expensive than other types of RAM
  - Commonly used in small amounts, such as in cache memory

Dynamic RAM (DRAM)

- ✔ DRAM is the most widely used type of RAM that constantly refreshes its data.
  - Holds data only for a short time
  - Requires continuous refreshing
  - Slower than SRAM
  - Less expensive and used as the main memory in most computers

DDR RAM (Double Data Rate RAM)

- ✔ DDR RAM is an improved version of DRAM that operates at higher speed and efficiency.

Different versions of DDR RAM:

DDR1: Oldest and slowest version
DDR2: Faster and more efficient than DDR1
DDR3: Common in older computers and laptops
DDR4: Very fast and widely used in modern systems
DDR5: Latest and fastest version designed for advanced computers

Memory Unit

Secondary Memory

✔ **Secondary storage** refers to non-volatile storage devices that hold data permanently.
✔ keeps data for a long time without the need for a continuous power source.

Importance of Storage Devices

- Provide permanent storage for data
- Allow storage of large volumes of information
- Serve as a backup solution
- Enable data recovery in case of system failures
- Facilitate file sharing and data transfer



Categories of Secondary Storage Devices

Category	Examples	Speed	Portability	Use Case
Magnetic Storage	HDD, Tape Drives	Moderate	Low	Long-term storage
Solid-State Storage	SSD, USB Drives	High	High	Fast data access
Optical Storage	CDs, DVDs, Blu-rays	Low	Moderate	Media storage
Cloud Storage	Google Drive, Dropbox	High	Very High	Remote backups

Memory Unit

Secondary Memory (HDD)

✔ A hard disk drive is used to store a large amount of data.
✔ HDD is a **type of magnetic storage device**.

Key Features of HDD:

•Stores large amounts of data (500 GB, 1 TB, or more)
•Non-volatile; retains data without power
•Contains moving parts (spinning disks)
•Uses magnetic storage technology
•Installed inside the computer case
•Slower than SSD
•Low-cost storage option



Memory Unit

Secondary Memory (SSD)

✔ A solid-state drive stores data in a fast way.
✔ It has no moving parts. It works silently and quickly.

Key Features of SSD:

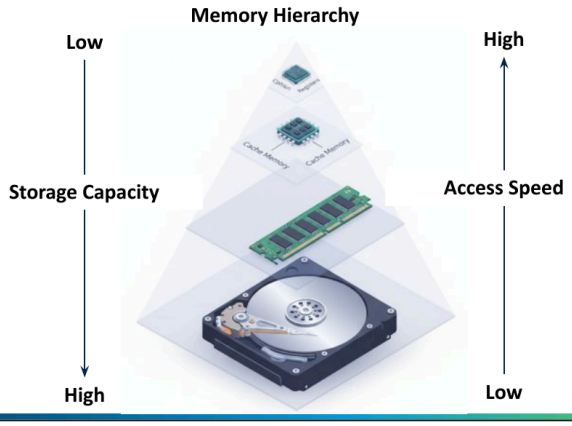
- Stores large data (256 GB, 512 GB, 1 TB)
- Non-volatile; retains data without power
- Uses flash memory for storage
- Faster than HDD
- Energy-efficient; saves laptop battery
- Lightweight and compact
- Silent operation (no moving parts)



Memory Unit

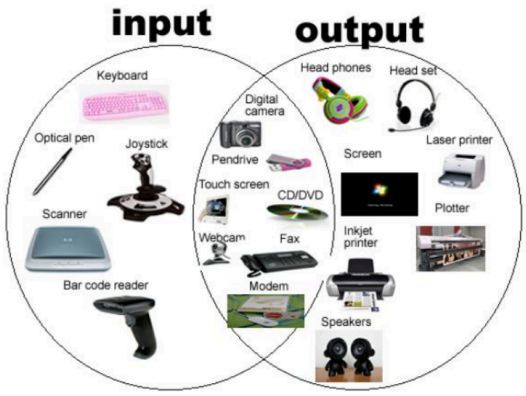
Register

- ✔ are a type of computer memory built directly into the processor or CPU (Central Processing Unit)
- ✔ is used to store and manipulate data during the execution of instructions.
- ✔ stores copies of frequently used data and instructions so that the CPU can access them much more quickly than if it had to fetch them from main memory (RAM) or from disk.



Input/Output Devices

Input Device: Hardware used to enter data into a computer, e.g., keyboard, mouse.
Output Device: Hardware that displays or produces the result of processed data, e.g., monitor, printer.



Assignment: Input and Output Devices

Objective:
Understand the different types of input and output devices, their functions, working principles, and importance in computer systems.

Instructions:

1. List at least **5 input devices** and **5 output devices**.
2. For each device, provide the following details:
    1. **Device Name**
    2. **Type** (Input/Output)
    3. **Function / Purpose**
    4. **How it works** (brief explanation)
    5. **Example Usage** (where it is commonly used)
    6. **Additional Info** (optional, e.g., advantages, limitations, special features)

Submission deadline: